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Field dependence-independence mediates the association between visual perception and mathematics. A cross-sectional study in children and preadolescents.

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ABSTRACT

Visuospatial abilities are associated with children's performance in mathematics. However, little is known about the contribution of visual perception, especially during preadolescence. Here, we aimed to test the relationship between visual perception, mathematics and field-dependent/independent cognitive style (FDI). We also focused on preadolescence, during which cognitive style tends to structure itself. Ninety-eight pupils were recruited in a public school in Rome (Italy) and divided into two age groups: children (age range = 6–10) and preadolescents (age range = 11–14). The relationship between visual perception, mathematics and FDI was tested by computing bivariate correlations as a preliminary step and using mediation analysis to test the mediating role of FDI. We found that FDI mediated this relation during preadolescence, enhancing the positive correlation between variables. In addition to the theoretical advances, the results of the present study may have an educational impact. Future studies should better investigate which types of training lead to mathematical achievement.

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KEYWORDS Visuospatial skills; mathematics; visual perception; field-dependence-independence cognitive style; mediation analysis; educational training

Introduction

Visuospatial abilities and mathematics

The study of visuospatial abilities (VSAs) has received increased attention in recent decades, especially due to the evidence that VSAs improve

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performance in science, technology, engineering and mathematics (STEM) (Young et al., 2018). VSAs involve a variety of different abilities, such as visual perception (Cui et al., 2017), spatial perception, mental rotation, and spatial visualization (Cornu et al., 2018). Nonetheless, most studies have focused on mental rotation: mental rotation is a predictor of children's performance in arithmetic operations, number line estimation, and mathematical problems between 6 and 9 years of age (Frick, 2019; Gilligan et al., 2019; Young et al., 2018) but not at 11 years of age (Mix et al., 2016). Even if little is known about the causal relation between VSAs and STEM, recent studies have suggested that visuospatial training with mental rotation impacts the general skills required to succeed in STEM in adults (Stieff & Uttal, 2015; Uttal et al., 2013) and children (Y. L. Cheng & Mix, 2014; Gilligan et al., 2020). In a few studies, it was found that training in mental rotation improved mathematical skills in first-, second- (Y.L. Cheng & Mix, 2014), and third-grade children in terms of missing problems (Gilligan et al., 2020). Additionally, training in spatial scaling improved number line estimation in third-grade children (Gilligan et al., 2020), and training in VSAs led to improved performance in mathematics at school in sixth-grade children (Lowrie et al., 2017). These studies pointed towards a key role of VSAs in mathematics. However, Hawes et al. (2015) did not find an effect of mental rotation training on mathematical skills; thus, any definite conclusions are still a matter of debate.

Visual perception and mathematics

Despite the evidence on the key role of VSAs, the involvement of the visual perception subcomponent received little attention until recent years. Thus, we focus on visual perception (VP) and its role in mathematics. VP is one of the components of VSAs, representing the starting point of visual processing (Cui et al., 2017). It encompasses the ability to identify features and details of a figure. Thus, VP plays a key role in mathematics, which is based on a symbolic system that requires fine visuospatial discrimination of different mathematical signs ($\pm$) and numbers (218/281). The normal acquisition and exercise of numerical skills and understanding make central use of several visual abilities, that is, perception of shape and orientation (e.g., 6 is different from 9), perception of position in right-to-left sequence (e.g., 218 and 281), and perception of vertical/horizontal alignment (e.g., during written arithmetic operations) (Giaquinto, 2007). For instance, to extract the information in a multi-digit

numeral, we need to recognize its constituent digits, which involves the perception of shape and orientation (Giaquinto, 2007). Thus, recognizing the Arabic code of numbers requires VP (Yu et al., 2022). In addition, VP reflects the ability to process relations between visual forms regardless of interference information, which is necessary to solve mathematical problems (Yu et al., 2022) and to visualize numerical relations in arithmetic tasks. In fact, several studies have shown a positive relation between VP and mathematical performance (e.g., Cui et al., 2017). According to a cross-sectional study, performances in a perception form task (that is, a geometric figure copying task), which requires the discrimination of detailed forms, are strongly related to mathematics, with the degree of association depending on the educational level of the children (Mix et al., 2016). However, a longitudinal study showed that children's visual form perception predicts mathematical achievement in multiple domains, such as arithmetic, geometry and statistics, 3 years later (Zhou et al., 2020). VP can contribute to lower-level maths categories (that is, numerosity comparison, digit comparison, and exact computation), which rely more on visual features, for students in primary (Zhou et al., 2015) and middle schools (Cui et al., 2017) but also to arithmetic fluency (Zhang et al., 2019). Indeed, Zhou et al. (2015) proposed the visual perception hypothesis to account for the association between numerosity processing (e.g., comparison of numbers of dots in two dot arrays) and computational fluency (e.g., solving of arithmetic tests). According to this hypothesis (Cui et al., 2017; Zhang et al., 2019; Zhou et al., 2015), VP could be the cognitive mechanism underlying the relationship between these two mathematical processes: both require the same cognitive processing of layout, symbols, and signs. This hypothesis is also supported by studies on children with developmental dyscalculia (DD), who have severe impairment in numerosity judgement (D. Cheng et al., 2018; Piazza et al., 2010; Zhou et al., 2015). Additionally, D. Cheng et al. (2020) developed a short-term numerosity training program for children with DD in Chinese primary schools. The findings indicated that numerosity training might enhance the arithmetic performance of children with DD, and mediation analysis revealed that this gain in arithmetic performance/training effect was totally mediated by an increase in VP. Taken together, these studies showed that VP can also be related to mathematics in the developmental age, although future studies are required to validate the causal role of VP in mathematics.

The effect of cognitive style

Interestingly, VP involves a variety of complex cognitive processes that vary among people and support different forms of conceptualization and learning. VP is crucial for pattern recognition (see Strasburger et al., 2011), and it is intrinsically related to field dependence/independence cognitive style (FDI) (Boccia et al., 2016). Indeed, this latter can be defined as ‘the extent to which the person perceives part of a field as discrete from the surrounding field as a whole, rather than embedded in the field; or the extent to which the organization of the prevailing field determines perception of its components (pp. 6–7)’ (Witkin, 1967; Witkin et al., 1977). The literature tends to describe FDI in more perceptual terms and effectively this cognitive style relies on overcoming the Gestalt principles of proximity, similarity, and good continuation (De Wit et al., 2017). Therefore, the ability to be field-dependent or independent is influenced by VP. For example, there is a line of research investigating the relation between FDI/EFT and visual acuity and showing a strong relation between them in individuals with autism spectrum disorder (Brosnan et al., 2012). In fact, FDI is usually measured using a version of the Embedded Figures Test (EFT; Witkin et al., 1971), in which a process of identifying the target within its context is required, which is known as disembedding. Individuals with greater accuracy or faster reaction times are described as being field independents (see the ‘Materials’ section). Specifically, whereas field dependence involves the tendency to accept the structure of the dominant field or context and the difficulty in separating an element from its context, field independence describes the ability to distinguish relevant from irrelevant items within a field and to provide structure in processing and reorganizing environmental information. Particularly, some research has shown that the individual disposition towards field independence is closely associated with several perceptual skills, but also with intellectual skills, including cognitive restructuring. These latter involve disembedding or flexibility of closure, which depicts the ability to extract salient information from the field, perspectivism, which describes the ability to recognize and adopt the other’s perspective concerning an array of objects (Witkin et al., 1977) and analytical intelligence (Raviv & Nabel, 1990). In fact, number of studies have shown that field independent and field-dependent individuals differ both in visuo-perceptual and spatial abilities, both in personal traits, social competencies, and vocational and occupational choices (Alamohodaei, 2002; Azari et al., 2013; Tinajero et al., 2011; Witkin et al., 1977). While field-dependent students may need more explicit instruction in problem-solving strategies or more exact definition of performance

outcomes than field-independent students, who may even perform better when allowed to develop their own strategies (Witkin et al., 1977). In particular, field-dependent students rely on the organization of the field as a whole, adopting a global perspective, and they are better when problems and tasks are already structured and analysed for them. By contrast, field-independent students are analytical and more capable of restructuring the perceptual field and imposing a structure if one is missing (Alamolhodaei, 2002). In addition, field-dependent individuals are more strongly influenced by the immediate social context and are more inclined to learn about social aspects of their environment, in fact they have a greater preference for humanistic professions, they learn social material more easily and show greater self-disclosure and cooperativeness. Conversely, field-independent individuals pay less attention to social problems and prefer professions that require high autonomous functioning and analytic thinking (Azari et al., 2013; Tinajero et al., 2011).

Yet, FDI, and specifically, field independence, also reflects the efficiency of different controlled mental processes, including executive functions such as cognitive flexibility, working memory, inhibition, and fluid intelligence (De Wit et al., 2017; Jia et al., 2014; Miyake et al., 2010), representing a critical factor in the analysis and restructuring of a given stimulus to solve a problem that requires the decontextualization of details (Pithers, 2002; Rémy & Gilles, 2014; Witkin et al., 1977).

Furthermore, FDI depicts a pervasive dimension of individual functioning, closely associated with several skills, such as spatial navigation (Boccia et al., 2016, 2017, 2019; Nori et al., 2021), spatial thinking (Rémy & Gilles, 2014), temporal processing (Teghil et al., 2019), semantic interference (Nappo et al., 2019), creativity (see Giancola et al., 2022b, 2022; see for a review; Giancola et al., 2022c), general cognitive abilities (Flexer & Roberge, 1980; Miyake et al., 2010) and achievements in academic fields (Tinajero & Paramo, 1998). Particularly, as concerns the education domain, a group of Asian researchers (Alamolhodaei, 2002; Azari et al., 2013) conducted several studies on the relationship between FDI and mathematical problem solving, finding that field-independent individuals perform better in problem-solving tasks. Indeed, field-independent students outperformed field-dependent students in both word problems (Alamolhodaei, 2002) and procedural problems (Azari et al., 2013), even after controlling for intelligence and mathematical anxiety.

A series of studies (e.g., Vaidya & Chansky, 1980; Van Blerkom, 1988; Witkin et al., 1977) demonstrated the association between FDI and

mathematics performance, suggesting that field-independent individuals outperform field-dependent ones due to their ability to develop analytical strategies, which allows them to reorganize and restructuring, in more rapid times, information to solve a mathematical problem. For instance, Onwumere and Reid (2014) found that field independence affected mathematics performance in students aged between 12 and 16. Additionally, it would seem that field independents did not use up working memory capacity with items not essential for the mathematical task. Accordingly, this tendency reduces the cognitive demands of working memory capacity and, in turn, enables higher mathematical performance.

The aim of the study

The aim of the present study is twofold. First, in line with previous studies in Western samples (Cui et al., 2017) demonstrating a positive correlation between VP and mathematics performance, we hypothesized that VP was associated with mathematics in the whole Italian sample as well. Second, the current study aimed to further investigate the relationship between VP and mathematics, addressing the potential impact of FDI, based on the aforementioned literature indicating the two-by-two relationship between VP, FDI, and mathematics. Specifically, we expected that FDI might be involved in the relationships between VP and mathematics, hypothesizing that field independence boosts this relation, particularly during preadolescence, when cognitive style exhibits an increasing trend. Indeed, in terms of developmental trends, some research has shown greater field independence with age (e.g., Akshoomoff & Stiles, 1996; Amador-Campos & Kirchner-Nebot, 1997), even though such a trend is not necessarily linear. For instance, no significant differences were found between age groups of 5–6 and 6–7 years, while a greater significant difference was shown between age groups of 7–8 and 8–9 years (e.g., Bigelow, 1971), and performance in FDI tasks tends to plateau around the age of 17 years (e.g., Amador-Campos & Kirchner-Nebot, 1997; Booth, 2006; Witkin, 1967). According to Witkin and Goodenough's (1981) view, learners tend to move to field independence by their mid-teenage years, with further learning having little subsequent effect (Pithers, 2002). In light of these assumptions, we performed a mediation model, in which VP was entered as the independent variable, mathematics (written operation) was the outcome, and FDI was the mediator.

Methods

Participants and procedure

Forty-two children and fifty-six preadolescents were recruited from a public institute in Rome (Italy). The research project was presented to the dean and to the teachers, and all of the students from the classes that expressed an interest in participating were included in the sample. All participants were Italian native speakers; none of them had primary visual or hearing impairments, had been diagnosed with a neurological condition, or had ever exhibited emotional or behavioural problems. Deficits in the general cognitive level were excluded by means of Raven's Coloured Progressive Matrices (Belacchi et al., 2008; Raven, 1986). Three participants (two children and one preadolescent) were identified as multivariate outliers and were discarded from the analyses. The final sample consisted of 40 children (mean age = 9.28 years, SD = 1.54 years, 57.5% F) and 55 preadolescents (mean age = 12.35 years, SD = 0.88 years, 56.4% F). The study was approved by the local ethics committee of the Psychology Department of Sapienza University of Rome, in accordance with the Declaration of Helsinki. Written and informed consent was obtained from the parents of each child participating in this research, and each student provided verbal assent to take part in the study. In addition, the pupils were informed that they could withdraw from the study at any time or ask for a break if they were too tired. Participants were tested individually in a quiet and well-lit room in their school and were asked to complete a battery of tests that measured FDI, VP and mathematical skills. These tests are validated and widely used in Italy. They are extensively described below. The order of the tests was randomized.

Materials

Test of Visual-Motor Integration (VMI; Beery & Beery, 2004, Italian version by; Preda, 2008). VMI is a paper and pencil test investigating visual-motor integration skills through the copy of a geometric design. It also includes two additional tests: VP and motor coordination. Given the goal of the current study, we used only the VP subtest. The VP is a figure matching task with 27 stimuli with a time limit and increasing difficulty: within 3 minutes, the participant must identify the identical shape to the geometric stimulus, which becomes increasingly complex among a series of alternatives (of 2 to 7). One point is awarded for each correct response, up

to a maximum of three consecutive incorrect answers. The total score is given by the sum of the correct responses (0–27).

The Children's Embedded Figure Test (CEFT; Karp & Konstadt, 1971; Italian version by ; Fogliani et al., 1984). The CEFT is a paper and pencil test used to assess children's predisposition towards FDI. Children were asked to find one of two simple coloured figures within a larger complex coloured figure (hereafter called a complex figure). After a training session in which children familiarized themselves with the simple figures and were trained to find them in the complex ones, they were presented with the 25 testing items in which they had to find the simple figure, namely, the tent (11/25) or the house (14/25). Different from the EFT (see below), there was no time limit, and the CEFT score corresponded to the number of correct responses (Minimum = 1; Maximum = 25). However, in our study, we considered the total number of errors, to avoid inverse scores for children and adolescents (see the paragraph below for the Embedded Figure Test, we used to test FDI in adolescents). Thus, lower scores indicated individuals with higher predisposition towards the field independence.

The Embedded Figure Test (EFT; Witkin et al., 1971, Italian version, Series A; Fogliani et al., 1984). In the present study, we tested preadolescents' FDI by using EFT, which is a paper and pencil test consisting of two sets of 12.9 × 7.7 cm cards, including 12 cards with complex-coloured geometric configurations (hereafter called complex figures) and 8 simple black/white shapes (hereafter called simple figures). The task requires finding a simple figure embedded in a complex figure as quickly as possible. The experimenter presented each complex figure for 15 s, during which participants were asked to orally describe the figure. Then, the experimenter removed the complex figure and presented the simple figure. After 10 s, the experiment removed the simple figure and showed the complex figure again, asking participants to find the simple figure embedded in the complex one. Following the standard administration procedure (Witkin et al., 1971), participants were explicitly required to advise the experimenter as soon as they found the simple figure and then to trace it using a stylus. The experimenter clocked the time. When a participant said that he or she had found the simple figure, the time spent since the second presentation of the complex figure was recorded: if the participant provided the correct response, that time represented the response time; otherwise, if the response was wrong, negative feedback was provided to the participant, and the experimenter continued to clock the time until the participant produced the correct response or until 180

s had passed. Response times were summed and then divided by the number of items (12) to compute the mean response time. Thus, in this sample, lower scores indicated individuals with a higher predisposition towards field independence.

Calculation Skill Assessment Test (AC-MT 6–11, Cornoldi & Cazzola, 2002; AC-MT11–14; Cornoldi & Cazzola, 2004). The AC-MT is a paper and pencil battery of mathematical tests consisting of several subtests to evaluate different mathematical competencies. According to the literature (e.g., Cui et al., 2017), we only used the ‘written operations in the classroom’ and “carry out the following operations” subtests, for the primary and secondary class, respectively (hereafter written calculation). Both of them consist of 8 mathematical operations to solve without temporal constraint (2 additions, 2 subtractions, 2 multiplications, 2 divisions), with the exception of the first class of the primary that requires the solution of 4. All of the operations are appropriate for the educational level attained. Each successful operation earns one point. To compare performances in the children’s sample, the raw scores were converted to proportions.

Statistical analysis

The data were analysed using IBM SPSS Statistics version 24 for Windows (IBM Corporation, Armonk, NY). Descriptive statistics were used to analyse the demographic features of the sample, and bivariate correlations were computed for preliminary analysis of the relation among all variables, whereas the hypothesis testing on the mediating role of FDI in the association between VP and written calculation was tested using the PROCESS macro for SPSS (version 3.5; Hayes, 2017). Note that we specifically used a mediation model since the focus of the current research is addressing the chain effect involving VP, FDI and written calculation. Figure 1 shows the theoretical model advanced in the current research for both children and adolescents (Model 4; Hayes, 2017). The significance of the mediating effects was analysed using 5,000 resamples of bootstrapped estimates with 95% bias-corrected confidence intervals (CIs) (Preacher & Hayes, 2008). Bootstrapping is a non-parametric approach enabling an accurate test of the mediating or moderating effect even in small- to medium-sized samples (e.g., Giancola, 2022; Giancola et al., 2022a, 2022d). The 95% CIs must not cross zero to satisfy the criteria of mediation (Preacher & Hayes, 2008). All significance was set to $p < 0.05$.

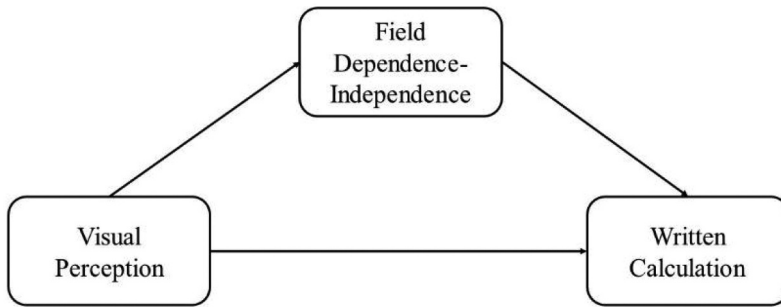


Figure 1. The theoretical model advanced in the current research.

Results

Table 1 shows means, standard deviations, and Pearson's correlations among the variables under investigation for both children and preadolescents.

Correlational analysis for children revealed that FDI correlated positively with VP ($r = -0.60$, $p < 0.01$) but not with written calculation ($r = 0.00$, $p > 0.05$). Moreover, no significant correlations were found between written calculation and age ($r = -0.03$, $p > 0.05$) or gender ($r = -0.22$, $p > 0.05$). Concerning preadolescents, the results indicated that VP correlated negatively with FDI ($r = -0.38$, $p < 0.01$) and positively with written calculation ($r = 0.27$, $p < 0.05$). Furthermore, FDI correlated negatively with written calculation ($r = -0.33$, $p < 0.05$), suggesting that the higher the disposition towards field independence (low response times), the higher preadolescents' ability in written calculation. Last, the written calculation was not found to be associated with age ($r = -0.22$, $p > 0.05$) or gender ($r = 0.24$, $p > 0.05$). In contrast to children, these results highlighted a potential mediating effect of FDI in the association between VP and written calculation.

Based on correlations, we performed a mediation analysis in the group of preadolescents. The results showed that VP was negatively associated with FDI ($B = -0.36$, $p < 0.01$), which, in turn, affected written calculation ($B = -0.04$, $p < 0.01$). Note that the residual direct effect was not significant ($B = 0.02$, 95% CIs $[-0.01, 0.05]$), whereas the indirect effect was significant ($B = 0.02$, 95% CIs $[0.01, 0.04]$). The total effect was also significant ($B = 0.04$, 95% CIs $[0.01, 0.07]$). Overall, these findings showed that FDI was involved in the association between VP and written calculation, fully mediating such a link.

Table 1. Means, standard deviations, and Pearson's correlations among all variables.

	Children (N = 40)					Preadolescents (N = 55)								
	M	SD	1.	2.	3.	4.	5.	M	SD	1.	2.	3.	4.	5.
1. Age	9.28	1.54	1					12.35	0.88	1				
2. Gender			-0.20	1						-0.09	1			
3. VP	0.80	0.09	0.44**	-0.08	1			23.78	2.36	0.09	-0.15	1		
4. FDI†	20.93	4.05	0.70**	-0.30	-0.60**	1		5.60	2.35	-0.25	0.10	-0.38*	1	
5. Written calculation	0.83	0.13	-0.03	-0.22	-0.23	0.00	1	5.13	2.33	-0.23	0.24	0.27*	-0.33*	1

†FDI was evaluated in terms of errors in children and response times in preadolescents.

Discussion

The key role of VSAs in mathematics has already been demonstrated in the literature, but relatively few studies have investigated the involvement of the VP in this process.

Over the last decade, a group of Chinese researchers investigated the role of VP, starting from the assumption that mathematics is based on a symbol system that requires fine visuospatial discrimination (see 'The visual form-perception hypothesis' in Zhou et al., 2020), especially by research focusing on the role of VP as a critical factor for approximate and arithmetic computation (D. Cheng et al., 2020; Cui et al., 2017; Zhou et al., 2015, 2020). Thus, Cui et al. (2017) revealed that, in addition to basic mathematical skills, exact computation was related to VP, making a significant contribution to mathematical achievement. Our results in a Western sample expand previous findings demonstrating a specific correlation between VP and mathematics: better performance in the VP task correlated with mathematical calculation success. To our knowledge, this is the first study replicating previous results by Cui et al. (2017) in a sample of preadolescents' Western countries. Thus, it is the first seminal evidence that the relation between VP and mathematics can be generalized and is not due to an overlearning of VP related to the Chinese logographic system (see Cui et al., 2017).

The main contribution of our study was to include cognitive style in this relation. In the literature, several studies have supported the strong relation between FDI and mathematics. Positive associations between field independence and academic success, such as in mathematics, were found in children (Morris et al., 2019) and adolescents (Alamolhodaei, 2002; Azari et al., 2013; Tinajero & Paramo, 1998): FI students tend to perform better in mathematics than FD students. We hypothesized that FDI, as an innate factor of individual predisposition linked to both the other two dimensions, could be the mediator of the relation between VP and mathematics. In line with our prediction, we found that FDI mediates this relation in preadolescence, enhancing its positive correlation: as field independence increases, the strength of the relation between VP and mathematics grows.

One may wonder whether the different pattern of results we found in children and adolescents may be due to different levels of practice with the tasks tapping mathematics, according to the results of Mix et al. (2016). These authors found that the relation between VP and mathematics increased as a function of the familiarization with the request of

the maths task. If this were the case, we should observe the same pattern of results in childhood and adolescence, since the two groups were tested with tasks that had the same level of familiarity/novelty for their grade level. However, this was not the case, and we should assume that the different patterns of results we found here support the idea that FDI may have a different role in the lifespan. Research has consistently shown a developmental change from field dependence to field independence in children and adolescents aged 3–17 years (Amador-Campos & Kirchner-Nebot, 1997).

Overall, our findings replicate previous findings in a preadolescent sample, introducing FDI as a relevant intervening variable. Importantly, the present results may have educational implications for school programs. FDI may be considered in adapting learning strategies and teaching modalities to improve the approach to mathematics. Indeed, in recent decades, a line of research has emerged focusing specifically on the adaptation of teaching methods to cognitive style, which provided data with potentially interesting implications in educational practice (Tinajero et al., 2011). Based on previous findings, for the early stages of mathematics learning, it is thought that field-dependent students may prefer tasks that are more structured or with social origin information (as occurs in group work) and tests such as closed-ended questions, whereas field-independent students may prefer unorganized materials and work independently to restructure them according to their needs. Bien (1974; Tinajero et al., 2011) found an improvement in the performance of field dependent primary school students when maths problems are structured by circling the relevant material. The improvement is such that their performance no longer differs from that of the field-independent students. Furthermore, some researchers (Threadgill-Sowder et al., 1985) have demonstrated that solving a drawn format of a mathematical problem is more helpful to field-dependent students than solving problems in other formats (verbal and telegraphic). Field-dependent students could benefit from the additional organizational features offered in a drawn format, whereas this one made little difference to field-independent students.

Our findings have an impact on didactics by suggesting the use and importance of integrated teaching that takes into account the differences between children and preadolescents to enable all of them to achieve scholastic success in the way that is most appropriate for them, adapting the scholastic means to the specific talents of each one.

Implications, future directions and limitations

Some limitations should be recognized. First, we provided evidence on the mediating effect of FDI in the association between VP and written calculation using a cross-sectional design, which does not allow for making cause and effect inferences. Future research should confirm our results by longitudinal research design. Also, although we used the bootstrapping method, which enables an accurate evaluation of the indirect effect in small-to medium-sized samples, our sample size is relatively small to test a more fine-grained timing of the relation between VP, written calculation and FDI, especially within the same years of age. Consequently, the sample size limits the possibility of differentiating development trajectories of the relation between VP mathematics and FDI. An English study (Morris et al., 2019) found that the association between FDI and mathematics was weaker in younger children (5–6 years) than in older children (9–10 years). According to the authors, this may suggest that younger children do not engage similar cognitive processes when completing these two activities, or it may be driven by a lack of variance in numerical operation scores in that year. In this light, we suggest that crucial changes in the development of FDI during childhood (see Amador-Campos & Kirchner-Nebot, 1997) affect the association between mathematics and VP in our study.

However, more research is needed. Training research could explore and verify whether there are some advantages in structuring mathematical materials relying on FDI. Ascertaining this hypothesis allows for deriving recommendations and guidelines for improving mathematical learning. In general, future research could systematically investigate how much this mediation effect of FDI on VP and mathematics impacts learning and student careers. Additionally, multicentre studies would confirm our findings in larger and more representative samples, allowing us to explore whether (and which) other factors are involved (e.g., socioeconomic and cultural conditions, gender, etc.). Furthermore, in this study, we used a single mathematical measure (written operation), which allows only a slight evaluation of mathematics. Accordingly, future research should confirm the results of the current study through a more comprehensive approach to mathematical abilities, employing different measures, such as numerical knowledge, mathematical problem-solving, and so forth, in order to better adapt the educational approach.

Conclusions

In conclusion, this is the first evidence of the role of FDI in mediating the positive relation between VP and mathematics, namely, as an arithmetical operation, in preadolescents but not in children. Future studies should better investigate which domains of mathematics are affected by FDI to design training leading to mathematical achievement.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The datasets generated and/or analysed during the current study are available from the corresponding author on reasonable request.

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